INCH-POUND
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DETAIL SPECIFICATION SHEET

ELECTRODES – WELDING, BARE, SOLID; AND FLUXES, SUBMERGED-ARC WELDING, CARBON AND LOW-ALLOY STEEL

This specification is approved for use by the Naval Sea System Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-23765.

1. SCOPE

1.1 Scope. This specification covers carbon and low-alloy steel, bare, solid welding electrodes, and fluxes for use with submerged-arc welding (SAW).

1.2 Classification. The classification system for the wire and flux is in accordance with AWS A5.17/A5.17M for carbon steel and A5.23/A5.23M for carbon-molybdenum steel except for the use of the prefix "MIL-" (e.g., MIL-EL12 for wire and MIL-F6A2-EL12 for flux).

1.2.1 Electrodes. The specified electrodes are EL8, EL12, EM12, or EM12K in accordance with AWS A5.17/A5.17M, and EA1 and EA2 in accordance with AWS A5.23/A5.23M; and supplied in forms 3b, 3c, 3d, 3e, and 4 in accordance with MIL-DTL-23765, and table I and 3.3 herein.

TABLE I. Form, size, and weight.

Form [1/]	**Weight (pounds)** [2/]
3b	25, 30, 35, 44, 50, 60, 250, 300, 600, or 750
3c	25, 50, or 60
3d	60 or 65
3e	150, 200, or 250
4	As specified by the purchaser

NOTES:

1/ Electrodes are furnished in diameters 1/16, 5/64, 3/32, 1/8, 5/32, 3/16, 7/32, and 1/4 inch.

2/ Tolerance on net weight is ±10 percent.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.2.2 Fluxes. Fluxes may be neutral or active, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-23765	-	Electrodes and Rods – Welding, Bare, Solid and Alloy-Cored, General Specification for
MIL-S-24238	-	Steel Plate, Carbon and Low Alloy

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS A5.17/A5.17M	-	Specification for Carbon Steel Electrodes and Fluxes for Submerged Arc Welding
AWS A5.23/A5.23M	-	Specification for Low-Alloy and High Manganese Steel Electrodes and Fluxes for Submerged Arc Welding

(Copies of these documents are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM E350	-	Standard Test Methods for Chemical Analysis of Carbon Steel, Low-Alloy Steel, Silicon Electrical Steel, Ingot Iron, and Wrought Iron

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Electrodes and neutral and active granular flux furnished under this specification shall be in accordance with MIL-DTL-23765 and as specified herein.

3.2 Qualification. Electrodes and fluxes furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.2 and 6.3).

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3.3 MIL-types.

3.3.1 Basic MIL-type. Basic MIL-type electrodes may have either a clean, bright finish or a uniform continuous well-bonded smoothly drawn copper coating on a clean surface. Diameters 3⁄32 inch and smaller may be coated with other types of rust preventatives, provided such coatings do not impair usability of the electrodes or the quality or soundness of weld metal deposits.

3.3.2 Special MIL-type with suffix "RC". Special MIL-type electrodes, designated by the suffix "RC", shall not be copper coated but shall have either a bright finish or a special protective coating, provided the coating does not impair usability of the electrode or the quality and soundness of the weld metal deposits.

3.4 Chemical composition. The chemical composition of the electrodes shall be in accordance with AWS specification electrodes indicated with the same classification without the prefix "MIL-".

3.4.1 Special MIL-type with suffix "RC". Special MIL-type electrodes that are not copper coated shall have copper and phosphorus content of 0.10 percent and 0.012 percent maximum, respectively.

3.5 Wall neutrality number (N). The neutral flux shall have a Wall neutrality number (N) of 35 or less. No testing shall be required for active fluxes to determine the neutrality number.

3.6 Mechanical properties. Unless otherwise specified (see 6.2), the mechanical properties of the weld metal resulting from wire and flux combination shall be in accordance with AWS A5.17/A5.17M or F7XX-ERXX-X of AWS A5.23/A5.23M.

3.7 Slag removal. The slag deposited during welding shall be capable of being readily removed from weld deposits with hand tools (not air or power operated).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examinations of MIL-DTL-23765 and the tests specified in table II.

TABLE II. Tests required for qualification inspection. 1/, 2/

MIL-types	Test	Procedure	Requirement
All electrodes Form 3b or 3d	Cast and helix	MIL-DTL-23765	MIL-DTL-23765
	Chemical	MIL-DTL-23765	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.
	Radiography	MIL-DTL-23765 3/	MIL-DTL-23765
	Tensile, bend, and impact 4/	MIL-DTL-23765 3/	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.
	Diffusible hydrogen 5/	MIL-DTL-23765	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.

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TABLE II. Tests required for qualification inspection – Continued. [1/, 2/]

MIL-types	Test	Procedure	Requirement
All fluxes	Wall neutrality number (N) [6/]	4.2.1.1 herein	3.5 herein
	Radiography	MIL-DTL-23765 [3/]	MIL-DTL-23765
	Tensile, bend, and impact [4/]	MIL-DTL-23765 [3/]	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.
	Diffusible hydrogen [5/]	MIL-DTL-23765	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.

NOTES:

1/ The mandatory strength and impact designators, and any optional designators in accordance with AWS A5.17/A5.17M and A5.23/A5.23M to be qualified, shall be the manufacturer's option.

2/ The tests shall be made in the as-welded condition.

3/ The weld test assembly shall be in accordance with AWS A5.17/A5.17M for carbon steel electrodes and AWS A5.23/A5.23M for carbon-molybdenum steel electrodes.

4/ Tensile testing includes ultimate tensile strength, 0.2 percent offset yield strength, and percent elongation in 2 inches. Impact tests shall be performed when required for the AWS classification being qualified.

5/ Diffusible hydrogen testing shall be performed when required for the AWS classification being qualified.

6/ Wall neutrality number (N) shall be determined for neutral flux.

4.2.1 Fluxes. Fluxes shall be used along with electrodes to make a weld test assembly in accordance with AWS A5.17/A5.17M or AWS A5.23/A5.23M, as required.

4.2.1.1 Wall neutrality number (N) for neutral fluxes. Prior to welding the test plates for flux qualification, the Wall neutrality number (N) for neutral fluxes shall be determined in accordance with 4.2.1.1.1.

4.2.1.1.1 Determination of N. Determination of N shall be in accordance with the following:

a. Weld test pad (minimum size – 1 inch wide by 4 inches long by ⅝ inch high) shall be prepared on a steel test plate in accordance with MIL-S-24238 using the SAW process with the flux to be tested in combination with a ⅛-inch diameter electrode. Welding parameters shall be as follows:

Electrode stickout: 1 to 1½ inches

Welding current: 500±10 amperes, direct current electrode positive

Arc voltage: 28±1 volts

Travel speed: 15±1 inches per minute

Interpass temperature: 300 degrees Fahrenheit (°F) maximum

b. A second weld test pad shall be made using the same parameters except that the arc voltage shall be increased to 36±1 volts.

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c. The top surfaces of the weld pads shall be ground or machined to clean metal. Then samples for analysis shall be removed by machining. No weld metal shall be removed within ½ inch of the base plate. The samples shall be chemically analyzed for silicon and manganese in accordance with ASTM E350. The N shall be calculated by the following formula using the absolute values of the change in weight percentage of silicon and manganese levels:

$$N = 100[|\Delta Si| + |\Delta Mn|]$$

4.3 Conformance inspection. Conformance inspection shall include the examinations of MIL-DTL-23765 and the tests specified in table III.

4.3.1 Certification of conformance. The certification of conformance shall be in accordance with MIL-DTL-23765.

TABLE III. Tests required for conformance inspection. 1/, 2/

MIL-types	Test	Procedure	Requirement
All electrodes All forms and sizes	Cast and helix	MIL-DTL-23765	MIL-DTL-23765
	Chemical	MIL-DTL-23765	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.
	Radiography	MIL-DTL-23765 3/	MIL-DTL-23765
	Tensile and impact 4/	MIL-DTL-23765 3/	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.
	Alloy identity	MIL-DTL-23765	MIL-DTL-23765
	Diffusible hydrogen 5/	MIL-DTL-23765	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.
	Plastic spool burn test 6/	MIL-DTL-23765	MIL-DTL-23765
All fluxes	Radiography	MIL-DTL-23765 3/	MIL-DTL-23765
	Tensile and impact 4/	MIL-DTL-23765 3/	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.
	Diffusible hydrogen 5/	MIL-DTL-23765	For carbon steel electrodes: AWS A5.17/A5.17M. For carbon-molybdenum steel electrodes: AWS A5.23/A5.23M.

NOTES:

1/ The AWS classification shall be as specified (see 6.2).

2/ The tests shall be made in the as-welded condition.

3/ Weld test assembly shall be in accordance with AWS A5.17/A5.17M for carbon steel electrodes and AWS A5.23/A5.23M for carbon-molybdenum steel electrodes.

4/ Tensile testing includes ultimate tensile strength, 0.2 percent offset yield strength, and percent elongation in 2 inches. Impact tests shall be performed when required for the AWS classification.

5/ Diffusible hydrogen testing shall be performed when required for the AWS classification.

6/ The plastic spool burn test shall be in accordance with MIL-DTL-23765.

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4.4 Base metal. Unless otherwise specified (see 6.2), the base metal steel used for the weld test assemblies shall be in accordance with AWS A5.17/A5.17M and A5.23/A5.23M.

4.5 Inspection of packaging and marking. Inspection of packaging and marking shall be in accordance with MIL-DTL-23765.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the welding materials described in 6.1.1 through 6.1.7 that, when deposited on the applicable base metals specified in 4.4, will meet the mechanical properties specified herein in the as-welded condition. Current stock of electrode types specified in this specification may be used until depleted; future requisitions for electrode types specified in this specification should refer to the current version of this specification.

6.1.1 MIL-EL8. MIL-EL8 electrodes are intended for making welds where maximum ductility or minimum alloy content in the weld is desired and is especially usable for heavy fillets and other welds subject to large shrinkage stresses.

6.1.2 MIL-EL12. MIL-EL12 electrodes result in higher strength than MIL-EL8 electrodes and are intended for welding carbon steel.

6.1.3 MIL-EM12. MIL-EM12 electrodes result in higher strength than MIL-EL12 electrodes and are intended for high speed welding of carbon steel.

6.1.4 MIL-EM12K. MIL-EM12K electrodes have better deoxidation properties than MIL-EM12 electrodes and are used for high speed welding of carbon steels.

6.1.5 MIL-EA1. MIL-EA1 electrodes result in higher strength than MIL-EM12 electrodes and are intended for welding of low-alloy steels.

6.1.6 MIL-EA2. MIL-EA2 electrodes result in higher strength than MIL-EA1 electrodes and are intended for welding of low-alloy steels.

6.1.7 Fluxes. Neutral fluxes covered in this specification are intended for multiple pass welding. Active fluxes are intended for single pass welds.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Whether the flux to be used is neutral or active (see 1.2.2).
c. Mechanical properties of weld metal if other than as specified (see 3.6).
d. The classification in accordance with AWS A5.17/A5.17M or AWS A5.23/A5.23M (see note 1/ of table III).
e. Other base metal steel to be used in tests, if required (see 4.4).
f. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 23765 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

Active fluxes

Neutral fluxes

Wall neutrality number (N)

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activities:
Army – AR
DLA – GS

Preparing activity:
Navy – SH
(Project 3439-2022-003)

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